Supplementary Methods, Results, and Reproducibility

Accompanies: Five-Year Medical Expenditure Value of Depression Symptom Resolution and Implications for Outcomes-Based Credits

Analysis completed October 28, 2026 | All model estimates reflect the original empirical analysis; no new model results were generated to prepare this document.

eMethods 1. Data, Population, and Operational End Point

MEPS annual data comprised 2019-2024 full-year consolidated files: 2019 HC-216, 2020 HC-224, 2021 HC-233, 2022 HC-243, 2023 HC-251, and 2024 HC-256 (original distribution filenames h216dta.zip, h224dta.zip, h233dta.zip, h243dta.zip, h251dta.zip, and h256dta.zip). The original annual data were supplied in MEPS.zip. The four-wave data came from 2018-2021 and 2019-2022 special longitudinal panels (More MEPS.zip) with design data h36u24dta.zip. Each participant is keyed by DUPERSID and PANEL as required.[1,2,3,4]

Adults aged 18-85 years were eligible for annual expenditure modeling with PHQ-2 in 0-6, positive SAQ weight, required age/sex/race/poverty/coverage/region codes, and nonnegative spending. Two observations (one in 2019 and one in 2021) with region coded -1 were excluded by this validity rule; the final scripts do not preserve a separate pre-exclusion separation-model result. Education with negative or missing coding was retained as an unknown group. PHQ-2 >=3 defined a positive screen, not a clinical diagnosis. The annual negative-screen sample was not restricted to people with past depression or a documented remission. MEPS does not supply PHQ-9 remission or the CMS 12-month assessment window.[5,6,7]

Annual spending outcomes were total medical expenditures and separately Medicare-paid, Medicaid-paid, and private-insurer-paid amounts. For each payer model, the relevant sample was selected using ever-covered insurance indicators; domains are allowed to overlap. Medical spending was not limited to depression-related diagnoses. Each person-year's expenditures were multiplied by the 2024 CPI-U annual average divided by the observation-year CPI-U annual average. Future projections maintained 2024 real spending levels rather than adding assumed expenditure growth.[1,4]

eTable 1. Eligible Annual MEPS Person-Years by Calendar Year

Year	Full-year records	Eligible adult person-years	Positive PHQ-2 screens
2019	28,512	17,226	1,351
2020	27,805	14,024	1,194
2021	28,336	15,174	1,343
2022	22,431	10,632	828
2023	18,919	9,412	816
2024	19,140	11,019	1,072

The final annual sample had 77,487 person-years (6,604 positive). Annual analysis used the SAQWTxxF questionnaire-specific weights, pooled VARSTR and VARPSU design identifiers; all unique survey PSUs were retained in domain-based influence-function variance estimation.[4]

eMethods 2. Maryland-Style Two-Part Spending Models

For each payer p and person-year i, let Y_ip be nonnegative annual expenditures and D_i indicate PHQ-2 positive status. The marginal expenditure mean was modeled using two components: Pr(Y_ip>0 | X_i,D_i)=expit(X_i beta_p + b_p D_i) and E[Y_ip | Y_ip>0,X_i,D_i]=exp(X_i gamma_p + g_p D_i). The regression was fit by weighted binomial logistic generalized linear modeling and weighted Gamma log-link generalized linear modeling on positive expenditures. The product of fitted probabilities and conditional positive means produces the unconditional expected expenditure.[8,9,10,11,12]

Core adjustment terms (consistent with the Maryland expenditure method) were age centered at 50 and scaled by 20, its square, sex, race, poverty category, grouped education, insurance category, census region, year indicator, and PHQ-2 symptom state. Race was coded from the annual survey variable, poverty categories from POVCAT, education from EDUCYR, insurance from INSCOV, and geography from REGION. Physical comorbidity terms were not added; these had been deliberately excluded from the policy specification. Calendar year was set to 2024 in standardized predictions after inflation adjustment.[8,10]

For each payer, marginal standardization averaged two predictions over the weighted covariate distribution of PHQ-2-positive adults in the payer domain: (a) setting PHQ-2 status to positive for everyone, and (b) setting it to negative for everyone, with the remaining covariates unchanged. The annual contrast Delta_p = mean(mu_positive) - mean(mu_negative) is the annual symptom-state spending contrast. Unlike a within-person causal remission effect, this contrast uses negative-screen comparison observations who may not have had depression, a limitation not removed by covariate standardization.[13,14]

The influence-function variance accounted for variation in both fitted generalized linear models and the composition of the standardized target population. Scores were aggregated by original VARSTR x VARPSU; in each stratum, cluster deviations were multiplied by n_h/(n_h-1). The full pooled survey roster contained 395 PSUs in 105 strata (290 design degrees of freedom). For the main model, annual 95% confidence intervals were calculated as the estimate +/- 1.96 times the survey-design standard error. The independent PPML mean-spending model was appendix-only and did not determine the published rate.[4,13,15]

eTable 2. Annual Payer Samples, Standardized State Means, and Primary Model Results

Payer	N	PHQ-2+ n	Zero-cost n	Negative state, $	Positive state, $	Annual contrast (SE), $
All medical expenditures	77,487	6,604	8,398	$9,456	$16,202	$6,746 (SE $627)
Medicare	27,468	2,749	1,797	$10,259	$18,758	$8,499 (SE $972)
Medicaid	15,033	2,666	3,101	$5,814	$9,047	$3,233 (SE $572)
Private insurance	45,768	2,674	5,696	$5,877	$9,024	$3,147 (SE $592)

The positive and negative adjusted means are model predictions for the same covariate-standardized baseline-positive target population. Survey CIs for annual contrast were: all $5,517-$7,975; Medicare $6,594-$10,404; Medicaid $2,112-$4,353; private insurance $1,986-$4,308.

eMethods 3. Five-Year Symptom-State Transition Model

The special four-wave analysis required PHQ-2 scores from rounds 2, 4, 6, and 8, valid demographic/insurance variables, a positive four-wave SAQ weight, and nonnegative expenditure variables in all four annual waves. Those longitudinal expenditure variables were eligibility filters only; postremission expenditures from these panels were not used to estimate the headline annual cost contrast. The final eligible longitudinal population comprised 4,496 adults. A positive-state (P) transition was one from a current PHQ-2 score >=3. A screen-negative-with-history state (R) required current PHQ-2 <3 and any observed earlier positive score, not necessarily a new positive-to-negative change in the immediately preceding year. Across people and waves there were 1,505 eligible transitions of these two types among 13,488 adjacent-wave pairs; multiple transitions from a respondent were combined in a respondent-level influence function.[2,3,16,17,18]

Weighted transition estimates were a = Pr(P next year | currently screen-negative, any earlier positive) = 0.2373854 and b = Pr(P next year | currently screen-positive) = 0.4884412. The projected two-state matrix T = [[1-a,a],[1-b,b]] (state order R,P) was an assumption imposed after a hypothetical 12-month landmark, with one trajectory initialized at R and its comparator at P. The model contrast in expected positive-state probabilities in future year k is (b-a)^(k-1). This was not an observed five-year trajectory, and the R-to-P transition observations were not restricted to the first year after newly documented remission.[18]

With 3% annual real discounting at the midpoint of each future year, the five-year discounted exposure multiplier is F = sum from k=1 to 5 of (b-a)^(k-1)/(1.03)^(k-0.5) = 1.3017825757. The year ending before the 12-month remission landmark is

excluded. No separate mortality or insurance churn process was fit. The Markov model uses beginning-of-year state rewards for each future interval, so changes of symptom status within each interval are not observed.[19]

eTable 3. Longitudinal Transition Inputs and Projection

Input	Value
Complete four-wave adults	4,496
Current negative screen, any earlier positive	444
Eligible transitions from positive state	1,061
Pr(P next year | R now)	23.74% (SE 2.55 percentage points)
Pr(P next year | P now)	48.84% (SE 2.59 percentage points)
Discounted five-year state-year factor	1.301783

eTable 4. Discounted Symptom-State Exposure by Future Year

Year after 12-month assessment	Discounted difference in state-years	Share of five-year factor
1	0.985329	75.69%
2	0.240168	18.45%
3	0.058539	4.50%
4	0.014269	1.10%
5	0.003478	0.27%

Descriptive, overlapping two-year record check: among 2,430 PHQ-2-positive paired annual observations, 1,166 had a positive screen in the next year (weighted 46.35%), compared with 48.84% in the four-wave analysis. These pairs were assembled from the same pooled annual MEPS sources; 870 came from people who also met the four-wave analytic eligibility criteria. Consequently this was not an independent-sample or external validation. The two-year pairing also cannot identify previous positive history among all currently negative respondents.[4]

eMethods 4. Credit Formula, Uncertainty, and Threshold Probabilities

For each payer p, the unadjusted five-year value was V_p = Delta_p F. The fixed spontaneous-remission rate was s=0.53, from Whiteford et al. (2013), a primary-care untreated-depression estimate based on 19 studies. The standardized reference case rate was B_p = V_p (1-s). The three recognition percentages alpha in {1.00,0.75,0.50} were then applied to produce C_p(alpha)=alpha B_p. Each same-rate credit is independent of observed provider remission counts, intervention costs, and intervention effectiveness.[8,20,21]

The 53% factor is a standardized pricing rule and must not be interpreted as an empirically identified causal-attribution fraction among observed remitters. It represents estimated remission within 12 months among untreated primary-care patients with major depression; the study population and remission definition differ from MEPS PHQ-2 transitions. The same fixed factor was used for all payers, with no payer-specific spontaneous-remission estimates. Therefore the fixed 53% input is not a validated estimate of the intervention-attributable share of any particular observed remission cohort.[20]

The code drew 120,000 multivariate-normal replicates for the four annual spending contrasts, retaining estimated cross-payer survey covariance. A second bivariate-normal draw represented the two longitudinal transition rates, with each simulated rate clipped between 0 and 1. The two draws were treated as independent by setting cross-source covariance to zero even though respondents in the underlying MEPS sources can overlap; this covariance was not estimated. The five-year factor and rate were recomputed for each draw, and the 2.5th and 97.5th percentiles define conditional uncertainty

intervals. The fixed 53% factor, outcome mapping, causal identification, and projection structure were not included as uncertain parameters.[15,22]

For each alpha, the reported conditional threshold probability equals the fraction of simulated spontaneous-remission-adjusted valuations satisfying B_p(draw) >= max(0,C_p(alpha)). Because the 100% rate equals the point estimate, this probability is approximately 50% by construction. It is not the probability that a Medicare or accountable care payment arrangement produces positive net savings; such an estimate would need the counterfactual number of additional remissions and program-specific payment arrangements. At a 100% recognition level, near-50% exceedance follows mechanically from comparing draws against their own center, and does not identify payer financial breakeven.[22]

eTable 5. Complete Five-Year Fixed Case Rates and 95% Conditional Simulation Intervals

Population	Recognized	Fixed rate, $	95% lower, $	95% upper, $	Conditional threshold probability
All medical expenditures	100%	$4,127	$3,335	$4,978	50.2%
All medical expenditures	75%	$3,096	$2,502	$3,734	99.5%
All medical expenditures	50%	$2,064	$1,668	$2,489	>99.9%
Medicare	100%	$5,200	$4,005	$6,467	50.1%
Medicare	75%	$3,900	$3,004	$4,851	98.4%
Medicare	50%	$2,600	$2,002	$3,234	>99.9%
Medicaid	100%	$1,978	$1,282	$2,703	50.3%
Medicaid	75%	$1,483	$961	$2,027	91.8%
Medicaid	50%	$989	$641	$1,352	99.8%
Private insurance	100%	$1,926	$1,211	$2,664	50.0%
Private insurance	75%	$1,444	$908	$1,998	90.6%
Private insurance	50%	$963	$605	$1,332	99.6%

eTable 6. Five-Year Values Before and After Spontaneous-Remission Pricing Adjustment

Population	Annual difference, $	5-y multiplier	Unadjusted 5-y, $	After 53% adjustment, $
All medical expenditures	$6,746	1.30178	$8,782	$4,127
Medicare	$8,499	1.30178	$11,064	$5,200
Medicaid	$3,233	1.30178	$4,208	$1,978
Private insurance	$3,147	1.30178	$4,097	$1,926

eMethods 5. Model Validation and Reproducibility

Implemented code checks included uniqueness of annual person-year records, nonnegative payment outcomes, a $5-tolerance check that selected payer components did not exceed total spending, model-matrix rank, convergence, positive/zero spending support, Hessian condition number <1e11, nonnegative payable credit candidates, and exact 75%/50% arithmetic. The analytic derivatives of standardized predictions were spot-checked by finite differences for three selected coefficients per GLM component, rather than for every coefficient. Full survey variance used original strata and PSU rosters with zero-contribution domain PSUs. The final code excluded two annual observations with invalid region codes; it does not include the pre-exclusion regression or the historical separation diagnostic. These are implementation checks, not proof of unbiased effect estimation or out-of-sample predictive validity.[10,11]

A separately fitted PPML log-link mean model used the same annual observations, covariates, and standardization target as the primary two-part model. Its similar positive contrasts provided an alternate-specification diagnostic, not independent-data or causal validation. The five-year formula was checked against matrix multiplication with the estimated transition rates and 25 random rate combinations. The independently repeated primary run reproduced all 10 archived aggregate CSV/JSON outputs byte-for-byte. An earlier, separately reproducible prospective remitter-versus-persister analysis used 690 eligible positive-screen person-transitions (526 people) and measured expenditures after the assessment year; its Medicare annual difference was -$7,141 (95% CI, -$17,542 to $3,260). This imprecise discordant result was not used to construct the positive primary rate card and must accompany its actuarial interpretation.[11,23]

eTable 7. Alternative Single-Part Expenditure Model Check (Same Annual Data)

Payer	Primary two-part, $	Alternative PPML, $	PPML minus primary, $
All medical expenditures	$6,746	$7,222	$476
Medicare	$8,499	$8,071	-$427
Medicaid	$3,233	$3,345	$113
Private insurance	$3,147	$3,064	-$83

eTable 8. CPI-U Factors Used to Express Each Observation Year in 2024 Dollars

Observation year	Annual CPI-U	Multiplier to 2024
2019	255.657	1.22699
2020	258.811	1.21204
2021	270.970	1.15765
2022	292.655	1.07187
2023	304.702	1.02949
2024	313.689	1.00000

eDiscussion. Actuarial Interpretation and Scope

The final rate-card estimates are adjusted annual associations between current PHQ-2 screen states followed by a projected five-year state-difference calculation; they are not observed medical savings after clinical remission. The annual screen-negative regression comparison included respondents with no observed depression history. Marginal standardization to screen-positive profiles cannot remove this structural difference. A separate analysis of observed PHQ-2 positive-to-negative versus persistently positive transitions and subsequent expenditures found a Medicare annual contrast of -$7,141 (95% CI, -$17,542 to $3,260), rather than the positive cross-sectional state association used in the headline. This disagreement is consequential to policy, and neither PPML agreement with the primary cross-sectional model nor computational reproducibility resolves it.[14,20,22]

The model includes all medical expenditures (or amounts paid by each insurer) after the remission landmark in an expected-state valuation. It does not measure the cost of the intervention needed to produce remission, which is unknown by design. Later-year medical expenditures may include continuing treatment or maintenance; MEPS does not identify which future services belong to an unspecified intervention. Mortality, insurer changes, uncertainty in spontaneous remission, and residual comorbidity differences are not modeled explicitly. The displayed payer domains overlap and should not be summed.[19,21]

The proposed measurement mechanism in CMS care models would use actual Quality ID #370 clinical remission, with an eligible depression diagnosis, baseline PHQ-9 score greater than 9, and a PHQ-9 score below 5 at approximately 12 months. The present calculation cannot directly price that clinical event because PHQ-9 scores and precise assessment dates are

absent from MEPS. The analysis therefore establishes a provisional proxy case-rate methodology to be validated using linked clinical and Medicare claims data before actual financial reconciliation.[5,6,7]

From an actuarial standpoint, the main strengths are large national annual expenditure samples, payer-specific estimation, transparent standardized predictions, explicit post-landmark five-year valuation, and reproducible code. The principal unresolved question is whether adjusted differences between symptomatic and screen-negative adults represent downstream costs among the actually remitted subgroup. The independent PPML analysis supports the annual symptom-state association but cannot resolve this structural discrepancy.[24,25]

Clinical Measure Crosswalk: CMS Specification and MEPS Valuation Proxy

The policy proposal would use the CMS remission measure for payment qualification. The empirical valuation dataset approximates its symptom transition, but does not itself identify patients who satisfy every clinical criterion. The table below makes this separation explicit rather than implying that survey scores were exact MIPS quality reports.[5,6,7]

eTable 9. Clinical Outcome Requirements Compared With Available MEPS Data

Quality ID #370 clinical specification	MEPS research implementation
Documented major depression or dysthymia	Elevated PHQ-2 symptom screen; no requirement for verified clinical diagnosis in the primary model
Initial PHQ-9 or PHQ-9M >9	PHQ-2 >=3 at first observation; not equivalent diagnostic/severity information
PHQ-9 or PHQ-9M <5 at 12 months	Next annual PHQ-2 <3 in longitudinal transition data; symptom-screen resolution proxy
12-month reassessment within +/-60 days	Approximately annual survey waves; exact assessment dates not established
Eligible adult measure rate	Adults 18-85 years in MEPS with required complete analytic variables
Additional eligibility and exclusions	Full CMS measure exclusions not reproduced in MEPS valuation
Future economic horizon	Five modeled years after the remission landmark; preceding treatment year excluded

Reproducibility Materials and Exact Output Files

A companion reproducibility archive contains code/run_final.py (annual two-part models; standardization; state transitions; credits), code/enhanced_loader.py (longitudinal source loading), code/validate_final.py (independent PPML and projection audits), code/generate_report.py (source result report), original FINAL_REPORT.md and README.md, execution/rerun/validation logs, and all aggregated results files. No individual-level MEPS data are redistributed. Three original inputs are required to re-execute: MEPS.zip, More MEPS.zip, and h36u24dta.zip; the bundled README provides exact commands and dependencies.[26,27,28]

SHA-256 (MEPS.zip): 65b1e4bc71e105be5bcb1a65e18505921b0e7d3f133b5e7476b28daa363737a4

SHA-256 (More MEPS.zip): 15f0d113c190588c36cf707c59328f332699c60ad2f4ab0719887a1c82a70de2

SHA-256 (h36u24dta.zip): 7f7fb31b7912169a9baf0d99b2a65cd1d0d03afe6ba26f0c3316f8a7f85e3659

Published inputs include AHRQ full-year consolidated, Panel 23/24 four-year longitudinal, and HC-036 pooled variance files; methodological precedents include Maryland's approved diabetes valuation, CMS Quality ID #370, and peer-reviewed depression-screening, symptom-course, health econometrics, survey, and decision-modeling studies. Complete bibliographic references follow.[6,8,20,29,30]

eMethods 6. Evidentiary Basis of Inputs and Assumptions

Population and measures. Annual expenditures and coverage are taken from MEPS full-year consolidated files. PHQ-2 >=3 is a positive screening threshold, not a diagnostic determination; conversion from a positive screen to a negative screen does not establish PHQ-9 clinical remission.[1,5,6,7]

Spontaneous remission. The 53% value is the estimated 12-month remission proportion in untreated adult primary-care major-depression samples, not an estimate measured in MEPS or a proportion causally attributable to a provider. The research applied this value once as a fixed pricing discount.[20]

Costs and standardization. The original Maryland two-part design motivates the logistic and Gamma components. Predictive margins are averaged over the baseline PHQ-2-positive covariate distribution, and 2024-dollar inflation factors use the national CPI-U annual averages.[8,9,10,13,14,29]

Longitudinal horizon. Four-wave MEPS transitions determine symptom persistence and recurrence; the five-year horizon is projected from the 12-month landmark using an annual stationary two-state model, with future costs discounted at 3% using a midyear convention.[2,3,18,19]

Statistical uncertainty. Analytic design-based influence functions and 120,000 normal draws propagated observed sampling variation in annual state-spending contrasts and symptom-transition rates. Annual and longitudinal sources were assigned zero cross-covariance for simulation, even though underlying respondents can overlap. The fixed 53% adjustment, selection and unmeasured confounding, outcome misclassification, intervention effects, and uncertainty in the state-projection architecture were not modeled. These intervals and exceedance probabilities are not financial-breakeven estimates.[15,22,24]

Model check. PPML is a separate diagnostic comparison, not a second published case-rate estimator. Its citation concerns consistent mean modeling with zero-valued outcomes rather than causal identification.[11,23]

eTable 10. Source-to-Method Evidence Traceability

Empirical or policy element	Implemented in code	Supporting source
National annual files	Annual MEPS 2019-2024	1
Four-year transitions	Panels 23 and 24	2
Complex sample variance	HC-036 pooled PSU and strata	4
Symptom definition	PHQ-2 >=3, later <3	5
Clinical measure comparison	CMS PHQ-9 12-month construct	6
Expenditure family	Two-part logistic and Gamma/log	10
Standardization	Weighted marginal prediction	13
Inflation	Annual CPI-U to 2024	29
Transition projection	Two-state annual model	18
Discounting	3% real rate convention	19
Spontaneous-remission input	Fixed 53%	20
Uncertainty propagation	Survey covariance plus joint simulation	22
Independent model diagnostic	Single-part PPML	23
Federal precedent	CMMI-approved Maryland credit	30

eMethods 7. Reproducible Computing Environment

The companion package contains all analytical Python scripts, source-file checksums, aggregated output tables, and execution and validation logs. It excludes respondent-level MEPS records; users obtain public-use inputs from AHRQ and verify their files against the SHA-256 manifest. Run the analysis and validator with the commands printed in README_REPRODUCE.md.[26,27,28]

Package versions in the execution environment: NumPy 2.3.5; pandas 2.2.3; SciPy 1.17.0; statsmodels 0.14.6; patsy 1.0.2. These versions are recorded for transparency; rebuilding with identical or supported compatible versions should be checked against stored results before policy use.

Research-document provenance. OpenAI GPT-6 assisted with drafting, bibliography assembly, and analytical code development or review. The accompanying executable checks and source-output comparisons document computational validation, not an independent external statistical audit or peer review. Final author accountability, research ethics assessment, citation sign-off, and policy validation remain the responsibility of the submitting research team.

Supplemental References

1. Agency for Healthcare Research and Quality. MEPS HC-256: 2024 Full-Year Consolidated Data File Documentation. August 2026. Accessed October 28, 2026. https://meps.ahrq.gov/data_stats/download_data/pufs/h256/h256doc.shtml

2. Agency for Healthcare Research and Quality. MEPS HC-236: Panel 23 Four-Year Longitudinal Data File, 2018-2021. Accessed October 28, 2026. https://meps.ahrq.gov/data_stats/download_data/pufs/h236/h236doc.shtml

3. Agency for Healthcare Research and Quality. MEPS HC-245: Panel 24 Four-Year Longitudinal Data File, 2019-2022. Accessed October 28, 2026. https://meps.ahrq.gov/mepsweb/data_stats/download_data_files_detail.jsp?cboPufNumber=HC-245

4. Agency for Healthcare Research and Quality. MEPS HC-036: 1996-2024 Pooled Linkage Variance Estimation File Documentation. Accessed October 28, 2026. https://meps.ahrq.gov/mepsweb/data_stats/download_data/pufs/h036/h36u24doc.shtml

5. Kroenke K, Spitzer RL, Williams JBW. The Patient Health Questionnaire-2: validity of a two-item depression screener. Med Care. 2003;41(11):1284-1292. doi:10.1097/01.MLR.0000093487.78664.3C

6. Centers for Medicare & Medicaid Services. Quality ID #370: Depression Remission at Twelve Months. 2026 MIPS Clinical Quality Measure Specification. Accessed October 28, 2026. https://qpp.cms.gov/docs/QPP_quality_measure_specifications/CQM-Measures/2026_Measure_370_MIPSCQM.pdf

7. Kroenke K, Spitzer RL, Williams JBW. The PHQ-9: validity of a brief depression severity measure. J Gen Intern Med. 2001;16(9):606-613. doi:10.1046/j.1525-1497.2001.016009606.x

8. Maryland Health Services Cost Review Commission. Maryland Diabetes Incidence Outcome-Based Credit Methodology. Approved May 2, 2019. Accessed October 28, 2026. https://hscrc.maryland.gov/Documents/Modernization/OBC/Maryland%20Diabetes%20Incidence%20Outcome-Based%20Credit%20Methodology%20-%20approved.pdf

9. Manning WG, Duan N, Rogers WH. Monte Carlo evidence on the choice between sample selection and two-part models. J Econom. 1987;35(1):59-82. doi:10.1016/0304-4076(87)90081-9

10. Manning WG, Mullahy J. Estimating log models: to transform or not to transform? J Health Econ. 2001;20(4):461-494. doi:10.1016/S0167-6296(01)00086-8

11. Buntin MB, Zaslavsky AM. Too much ado about two-part models and transformation? Comparing methods of modeling Medicare expenditures. J Health Econ. 2004;23(3):525-542. doi:10.1016/j.jhealeco.2003.10.005

12. Manning WG, Basu A, Mullahy J. Generalized modeling approaches to risk adjustment of skewed outcomes data. J Health Econ. 2005;24(3):465-488. doi:10.1016/j.jhealeco.2004.09.011

13. Graubard BI, Korn EL. Predictive margins with survey data. Biometrics. 1999;55(2):652-659. doi:10.1111/j.0006-341X.1999.00652.x

14. Muller CJ, MacLehose RF. Estimating predicted probabilities from logistic regression: different methods correspond to different target populations. Int J Epidemiol. 2014;43(3):962-970. doi:10.1093/ije/dyu029

15. Lumley T. Analysis of complex survey samples. J Stat Softw. 2004;9(8):1-19. doi:10.18637/jss.v009.i08

16. Solomon DA, Keller MB, Leon AC, et al. Multiple recurrences of major depressive disorder. Am J Psychiatry. 2000;157(2):229-233. doi:10.1176/appi.ajp.157.2.229

17. Burcusa SL, Iacono WG. Risk for recurrence in depression. Clin Psychol Rev. 2007;27(8):959-985. doi:10.1016/j.cpr.2007.02.005

18. Sonnenberg FA, Beck JR. Markov models in medical decision making: a practical guide. Med Decis Making. 1993;13(4):322-338. doi:10.1177/0272989X9301300409

19. Sanders GD, Neumann PJ, Basu A, et al. Recommendations for conduct, methodological practices, and reporting of cost-effectiveness analyses: Second Panel on Cost-Effectiveness in Health and Medicine. JAMA. 2016;316(10):1093-1103. doi:10.1001/jama.2016.12195

20. Whiteford HA, Harris MG, McKeon G, et al. Estimating remission from untreated major depression: a systematic review and meta-analysis. Psychol Med. 2013;43(8):1569-1585. doi:10.1017/S0033291712001717

21. Counts NZ, Kuklinski MR, Wong V, Feinberg ME, Creedon TB. Comparison of estimated incentives for preventing postpartum depression in value-based payment models using the net present value of care vs total cost of care. JAMA Netw Open. 2022;5(4):e229401. doi:10.1001/jamanetworkopen.2022.9401

22. Briggs AH, Weinstein MC, Fenwick EAL, Karnon J, Sculpher MJ, Paltiel AD. Model parameter estimation and uncertainty analysis: a report of the ISPOR-SMDM Modeling Good Research Practices Task Force Working Group-6. Med Decis Making. 2012;32(5):722-732. doi:10.1177/0272989X12458348

23. Santos Silva JMC, Tenreyro S. The log of gravity. Rev Econ Stat. 2006;88(4):641-658. doi:10.1162/rest.88.4.641

24. Husereau D, Drummond M, Augustovski F, et al. Consolidated Health Economic Evaluation Reporting Standards 2022 (CHEERS 2022) statement: updated reporting guidance for health economic evaluations. BMJ. 2022;376:e067975. doi:10.1136/bmj-2021-067975

25. Vandenbroucke JP, von Elm E, Altman DG, et al. Strengthening the Reporting of Observational Studies in Epidemiology (STROBE): explanation and elaboration. PLoS Med. 2007;4(10):e297. doi:10.1371/journal.pmed.0040297

26. Seabold S, Perktold J. Statsmodels: econometric and statistical modeling with Python. Proceedings of the 9th Python in Science Conference. 2010. https://proceedings.scipy.org/articles/proceedings-2010.pdf

27. Virtanen P, Gommers R, Oliphant TE, et al. SciPy 1.0: fundamental algorithms for scientific computing in Python. Nat Methods. 2020;17(3):261-272. doi:10.1038/s41592-019-0686-2

28. Harris CR, Millman KJ, van der Walt SJ, et al. Array programming with NumPy. Nature. 2020;585(7825):357-362. doi:10.1038/s41586-020-2649-2

29. US Bureau of Labor Statistics. Consumer Price Index for All Urban Consumers: annual average indexes, 2019-2024. Accessed October 28, 2026. https://www.bls.gov/regions/mid-atlantic/data/consumerpriceindexannualandsemiannual_table.htm

30. Centers for Medicare & Medicaid Services, Center for Medicare and Medicaid Innovation. Outcome-Based Credit Methodology on Diabetes Incidence: Approval Letter. July 17, 2019. Accessed October 28, 2026. https://hscrc.maryland.gov/Documents/Modernization/OBC/CMS%20Appoval%20for%20Outcome-Based%20Credit%20Methodology%20on%20Diabetes%20Incidence%20%28Signed%29%20%281%29.pdf